

Apiwit Karnjanavivin

Email: apiwit.four@gmail.com | Phone number: +49 176 23820324 | Date of birth : 30 Oct 1997 | Nationality : Thai | Website : <https://misterfourxxx.github.io/portfolio>
Address : Brühlstrasse 43, 88299 Leutkirch im Allgäu, Germany | LinkedIn : www.linkedin.com/in/apiwit-karnjanavivin-36a72831b | GitHub : <https://github.com/MisterFOURXXX>

Education

Master of Science in Data Science, Artificial Intelligence, and Digital Business (M.Sc.) - First Class Honor June 2024 – September 2025
Gisma University of Applied Science
Potsdam, Germany
Final Dissertation: Better than it is: enhancing DeepSeekMoE architecture-based chatbots using dynamic routing algorithms

Bachelor of Engineering in Industrial Engineering (B.Eng.) August 2016 – April 2021
Kasetsart University
Bangkok, Thailand
Final Dissertation: Designing of lean-based hydroponic farm, its business plan, and financial analysis

Research Experience

Research Assistant - First author and main contributor (Volunteer) July 2025 – Present
Gisma University of Applied Science
Potsdam, Germany

- Proposed enhancing the DeepSeekMoE architecture by redesigning its routing algorithm, auxiliary load balance loss, and model training process to optimizing computational resource usage while retaining performance in training, fine-tuning, and inference.
- Co-authored a research paper targeted ACM publication, contributing introduction, foundations, related works, approach, and evaluation.
- Designed and executed ablation studies on the related routing algorithm, auxiliary load balance loss, and model architecture parameters on the prototype model with measurement metrics following research objectives to verify research claims and understand interactions between them.
- Contributing publication repository with ensure reproducibility for research paper publication of the baseline and prototype models with monitoring experiments, according to the objectives of the research.
- Collaborating on the research paper with a supervising professor through regular progress updates and iterative improvements based on professor’s feedbacks.

Professional Experience

In-Process Warehouse Leader October 2021 – November 2023
Raku Innovation Co., Ltd.
Bangkok, Thailand

- Directed Material Requirements Planning (MRP) and inventory audits across production warehouse and production lines using ERP systems (SAP B1).

Projects

- A Survey of LLMs Architectures**
- Explored and implemented widely used industrial LLM architectures—including GPT, Mixture of Experts (MoE), and Small Language Models (SLMs)—to study their internal mechanisms and implement on fine-tuning, text generation strategies, and model evaluation.
- A Survey of LLMs Fine-Tuning Approaches**
- Explored and implemented modern LLM fine-tuning methods to enhance model performance and reasoning, encompassing preference optimization, reward modeling, and reinforcement learning algorithms.
- A Survey of LLMs Fine-Tuning Optimization Approaches**
- Explored and implemented LLM fine-tuning optimizations and detailed their mechanisms on how does it can optimized computational resource and enhance model performance, including QLoRA, mixed precision, 8-bit optimizers, knowledge distillation, prompt tuning, and ZeRO-3.
- A Survey of Retrieval-Augmented Generation (RAGs)**
- Built and explored components of modern industrial RAG pipelines to reduce hallucination and enhance LLMs performance, including data ingestion, chunking, embedding, vector storage, retrieval, prompt engineering, conversational memory, and evaluation.
- Hospital Stay Prediction - Machine Learning Operations (MLOps)**
- Designed and deployed an end-to-end production MLOps pipeline using a stacked ensemble of machine learning models, covering the full lifecycle from training to production deployment, consisting of experiment tracking, hyperparameter optimization, data versioning, containerization, AWS deployment, and CI/CD pipelines.
- Fine Tuning LLM - Large Language Model Operations (LLMOps)**
- Developed a LLMOps pipeline for fine-tuning and deploying LLMs to production, using Direct Preference Optimization (DPO) and QLoRA, consisting of experiment tracking, data versioning, multi-metric evaluation, Dockerization, FastAPI inference, AWS deployment, and CI/CD pipelines.

Retrieval Augmented Generation (RAG) - Large Language Model Operations (LLMOps)

- Developed a production-grade Retrieval-Augmented Generation (RAG) pipeline to gain a more comprehensive understanding of RAG pipeline components for production, consisting of data ingestion, chunking, embedding, vector storage, retrieval, prompt engineering, conversational memory, model evaluation, and CI/CD pipelines.

Adversarial Text Generation Against LLM Judges

- Designed and developed an AI safety research framework that leverages adversarial text generation to enhance defense strategies for identifying vulnerabilities in LLM-as-a-judge decision committees from LLMs' attack.

Memory Poison Audit for long-term LLM memory threats

- Designed and built a research framework for developing defense mechanisms in long-term LLM memory in vector databases, using adversarial perturbation strategies and an anomaly-based retrieval sanitization layer to block context leakage without modifying core model weights and evaluated it using attack mechanisms.

eBPF Gradient Aggregation for Distributed Machine Learning

- Designed and developed a high-performance of distributed ML system applying eBPF and XDP kernel-bypass techniques to accelerate gradient aggregation evaluate communication latency, CPU utilization, and system scalability across custom protocol designs.

High-Throughput LLM Engine

- Designed and developed a research framework and system simulator for high-throughput LLM training and fine-tuning that applies mathematical latency models to evaluate collective communication backends (NCCL, XDP, Gloo), including infrastructure simulation, workload modeling, resource monitoring, and performance analysis.

Project documentation and code available at : <https://github.com/MisterFOURXXX?tab=repositories>

Achievements

- Social Media Extremism Detection Challenge - Top 24 percent of participants Kaggle Competition
- CIBMTR - Equity in post-HCT Survival Predictions Competition - Top 33 percent of participants
- PHEMS Hackathon: Pediatric Sepsis Prediction - Top 33 percent of participants
- Beyond Visible Spectrum: AI for Agriculture 2025 - Top 40 percent of participants
- FlightRank 2025: Aeroclub RecSys Cup - Top 45 percent of participants

Project documentation and code available at : <https://github.com/MisterFOURXXX/kaggle-competitions>

Training and Certifications

Operations (MLOps and LLMOps)

Coursera

- MLOps | Machine Learning Operations – Duke University
- Large Language Model Operations (LLMOps) – Duke University

Machine Learning, Neural Network, and Deep Learning

- IBM Machine Learning – IBM
- IBM AI Engineering – IBM
- Applied Machine Learning – Johns Hopkins University
- TensorFlow 2 for Deep Learning – Imperial College London
- IBM Deep Learning with PyTorch, Keras and Tensorflow – IBM
- Data Analytics and Deep Learning – Illinois Tech

Generative AI and Large Language Models (LLMs)

- IBM Generative AI Engineering – IBM
- IBM RAG and Agentic AI – IBM

Skills and Proficiencies

Programming Languages

- Python, SQL, Shell, C, C++, Go, Git, LaTeX (Overleaf)

Data Science and Visualization

- Polars, Pandas, NumPy, Plotly, Seaborn, Matplotlib

Machine Learning, Deep Learning, and Large Language Models

- Scikit-learn, PyTorch, TensorFlow/ Keras, Hugging Face, Unsloth, LangChain, LangGraph, vLLM

Systems, Infrastructure, and Operations

- MLflow, DVC (Data Version Control), Oracle Cloud, AWS, Terraform, Docker, Kubernetes, FastAPI

Languages

- English (Fluent), Thai (Native), German (Beginner)

References

Prof. Dr. Mohammad Mahdavi
mohammad.mahdavi@gisma.com

Gisma University of Applied Sciences, Professor

Prof. Dr. Mahmoudreza Babaei
babaei.mahmoudreza@gisma-education.com

Gisma University of Applied Sciences, Professor